

3D-Printed Prosthetic Foot Project

Problem

- Must withstand 150 lb heel loading at -15° and 50 lb toe loading at $+20^\circ$ while keeping maximum deflection under 2 cm, all within an 8–10 in footprint using only PLA
- The challenge was to design a lightweight, low-cost 3D-printed foot that satisfies bending, shear, and axial stress limits

Method

- Designed multiple iterations in Onshape, evolving from solid geometries to a material-efficient I-beam keel with a triangular support truss
- Modeled the foot as a cantilever beam under angled point loads to calculate bending, shear, axial stress, and deflection using mechanics of materials
- Optimized infill (15%) to minimize mass while maintaining structural integrity

Result

- Successfully supported 150 lb heel loading with a maximum deflection of 0.47 cm
- Achieved a factor of safety >70 for bending and axial stresses
- Produced a 177 g prosthetic foot at a material cost of \$17.71
- Identified failure at the ankle-foot interface during toe loading, informing clear design improvements for future iterations

